

Janhavi Wararkar

Junior Embedded Engineer

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SUMMARY

Junior Embedded Engineer with a B.Tech in Electronics and Telecommunication (Honors in Semiconductor Technology) from Pimpri Chinchwad College of Engineering, and eight months of hands-on experience at DAO EVTech building firmware, diagnostics tools and communication protocols for production electric vehicles.

EXPERIENCE

Junior Embedded Engineer | DAO EVTech Pvt. Ltd., Pune

July 2026 – Present

- Architected and led development of the vehicle's Distance-to-Empty (DTE) algorithm for production electric scooters, blending segmented historical consumption (1/5/15 km weighted windows) with real-time throttle-based power prediction (70/30 blend), SOC non-linearity and temperature compensation, and idle-drain modeling.
- Benchmarked the DTE engine against Ola, Bajaj, TVS and Ather approaches, reaching roughly $\pm 8\%$ average error, a 56% improvement over a simple moving-average baseline; served via a Flask/WebSocket backend to a live dashboard.
- Extended CAN bus diagnostics on production Ola S1 Pro Gen 2 two-wheelers (MoveOS 4/5), designing and running component-disconnect and component-swap tests across VCU, MCU and BMS CAN nodes with CANTest and USBCAN-2E analyzers to isolate inter-ECU fault behaviour.
- Led a root-cause failure analysis in Hall-effect sensor failures in a production EV BLDC hub motor: built an ESP32-based monitoring that showed peak condensation risk occurs just after motor shutdown, not during operation, informing Hall sensor PCB protection.
- Mentored two interns on embedded firmware development.

IoT Intern | DAO EVTech Pvt. Ltd., Pune

January 2026 – June 2026

- Built a production-vehicle datalogger capturing speed and throttle over the company's proprietary single-line bus, BMS data over RS-485 and vibration/temperature data; computed DTE on-device and logged it locally with GPS tracking, shown live on a real-time dashboard.
- Designed the complete vehicle wiring harness and validated it end-to-end through the team's in-house TMS (Test Management System) app for test tracking and assignment.
- Developed a BMS monitoring tool implementing the RS-485 read/control protocol (CRC-16-Modbus) to decode SOC, SOH, voltage, current, temperature and fault-status telemetry.
- Ran comparative firmware validation across BMS versions (reference vs. test units), identifying scaling, offset, SOH and temperature-reporting defects and documenting recommended firmware fixes.
- Reverse-engineered the CAN network of a production Ola S1 Pro Gen 2 by physically tapping VCU, MCU and battery nodes with CANTest and USBCAN-2E analyzers, documenting inter-ECU traffic and physical test constraints.

TRAINING PROGRAMS

VLSI Design Intern | Maven Silicon, Bengaluru

Jun 2025 – Jul 2025

- SPI RTL design in Verilog HDL using Xilinx Vivado, performed simulation, synthesis and verification in Xilinx Vivado.

Embedded System Developer Virtual Internship | AICTE / EduSkills, Microchip

Jan 2024 – Mar 2024

- 10-week Microchip University curriculum: Advanced and Embedded C, PIC16 architecture, MPLAB X IDE, MCC state machines, dsPIC motor control, BLE and Azure IoT sensor nodes.

EDUCATION

B.Tech, Electronics and Telecommunication | Pimpri Chinchwad College of Engineering, Pune

2022 – 2026

- CGPA 8.17 | Honors in Semiconductor Technology, CGPA 8.05

PROJECTS

Smart Transportation Hub with Load-Cell Weighing

- Designed an ESP32 vehicle tracking hub (GPS, IMU crash detection, Wi-Fi/GSM alerts) with a multi-load-cell weighing subsystem for total and distributed load, adding real-time overload and imbalance detection.
- **Skills:** ESP32 firmware, load-cell/HX711 signal conditioning, sensor fusion, GSM alerting

IoT Health Monitoring for Elderly and Soldiers over LoRaWAN

- Designed a low-power wearable that reads vital signs, environmental parameters and GPS location and sends them over LoRaWAN to a central system, with threshold-based alerts for abnormal vitals or conditions.
- **Skills:** LoRaWAN, low-power sensor interfacing, Embedded C
- **Outcome:** Long-range, battery-powered vitals link for areas without reliable Wi-Fi or cellular coverage

Voice-Based Smart Notice Board on Raspberry Pi

- Built a networked notice board that accepts voice commands and multimedia from a companion app, auto-generates notices using AI and renders them on LCD displays driven by a Raspberry Pi.
- **Skills:** Raspberry Pi Linux development, speech-command processing, display driving, app-to-device sync
- **Outcome:** Hands-free, remotely updatable notices with no manual on-site changes

SKILLS

- **Protocols:** CAN, RS-485/MODBUS, I2C, SPI, UART, LoRaWAN, BLE, MQTT
- **Hardware:** ESP32, Raspberry Pi 4B, STM32 (ARM Cortex-M), PIC16, 8051, Arduino
- **Languages:** C, C++, Embedded C, Python, Verilog HDL, SystemVerilog
- **Tools:** ESP-IDF/PlatformIO, STM32CubeIDE, MPLAB X IDE, Xilinx Vivado, CANTest, USBCAN-2E, SQLite, Git, EasyEDA Pro, MATLAB
- **Practice:** FreeRTOS firmware, peripheral drivers (ADC, PWM, Timer, GPIO), protocol reverse engineering, firmware validation and root-cause analysis, wiring harness design

CERTIFICATIONS

- Maven Silicon: VLSI SoC Design; RISC-V Processor, RV32I Base ISA
- Udemy: Mastering Microcontroller and Embedded Driver Development; Verilog HDL, VLSI Hardware Design Masterclass
- Microchip University: Advanced C Programming, Advanced Embedded C, PIC16 Architecture, BLE, dsPIC Motor Control